

Spectral analysis I

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University of Chicago

Department of Economics

Mikkel Plagborg-Møller

Motivation: Random sums of sinusoids

- Fix $N \in \mathbb{N}$ and define

$$\omega_j \equiv \frac{2\pi j}{N} - \pi, \quad j = 0, 1, \dots, N-1.$$

- Let the fct $s: [-\pi, \pi] \rightarrow \mathbb{R}_+$ satisfy $s(\omega) = s(-\omega)$, $\omega \in [0, \pi]$.
- Let $\{A(\omega_j), B(\omega_j)\}_{j=0,1,\dots,N-1}$ be $2N$ mean-zero, mutually uncorrelated random variables, such that

$$\text{Var}(A(\omega_j)) = \text{Var}(B(\omega_j)) = \frac{2\pi}{N} s(\omega_j), \quad j = 0, 1, \dots, N-1.$$

- Exercise:** Show that the scalar time series defined by

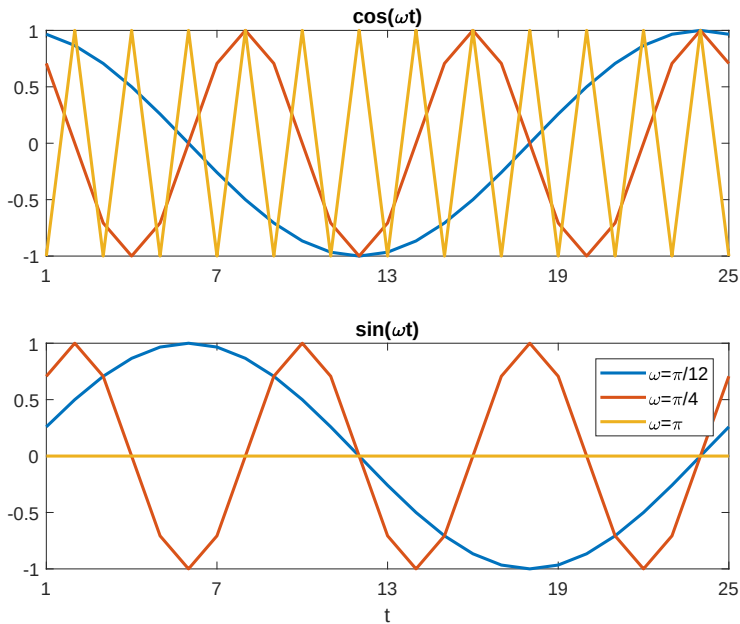
$$y_t = \sum_{j=0}^{N-1} \{A(\omega_j) \cos(\omega_j t) + B(\omega_j) \sin(\omega_j t)\} \quad \text{for all } t$$

is covariance stationary with

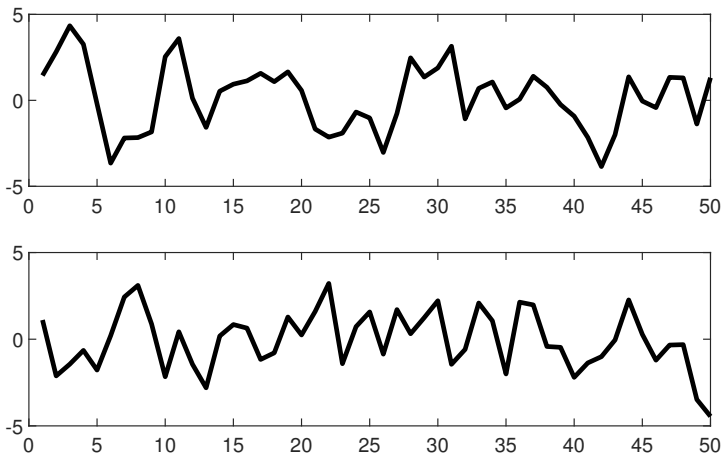
$$\mu_y = 0, \quad \Gamma_y(\ell) = \frac{2\pi}{N} \sum_{j=0}^{N-1} s(\omega_j) \cos(\omega_j \ell) \quad \text{for all } \ell \in \mathbb{Z}.$$

(Hint: $\cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\alpha - \beta)$.)

Motivation: Random sums of sinusoids (cont.)



Motivation: Random sums of sinusoids (cont.)



Two realizations of random sums of sinusoids with

$$N = 100, \quad A(\omega_j), B(\omega_j) \sim N\left(0, \frac{2\pi}{N}s(\omega_j)\right), \quad s(\omega) = 1 - \frac{|\omega|}{\pi}.$$

Motivation: Random sums of sinusoids (cont.)

$$y_t = \sum_{j=0}^{N-1} \{A(\omega_j) \cos(\omega_j t) + B(\omega_j) \sin(\omega_j t)\}, \quad \Gamma_y(\ell) = \frac{2\pi}{N} \sum_{j=0}^{N-1} s(\omega_j) \cos(\omega_j \ell).$$

- y_t is a random sum of sinusoids of different **frequencies** ω_j . The “variance attributable to frequency ω_j ” is $\frac{2\pi}{N} s(\omega_j)$.
- The slide appendix shows that, if N is odd, then for all m ,

$$s(\omega_m) = \frac{1}{2\pi} \sum_{|\ell| < N/2} \cos(\omega_m \ell) \Gamma_y(\ell) \approx \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} \cos(\omega_m \ell) \Gamma_y(\ell). \quad (*)$$

- **Spectral representation theorem** (loosely): Any scalar cov. stat. time series $\{y_t\}$ can be approximated arbitrarily well by a random sum of sinusoids as $N \rightarrow \infty$. (Details later.)
- Hence, the variance of y_t attributable to frequencies $[\omega, \omega + d\omega]$ is given by $s(\omega)d\omega$, with $s(\omega)$ as defined in equation (*).

Spectral density function

- In this slide deck we only consider scalar time series.

Definition

Let $\{y_t\}$ have absolutely summable ACF $\Gamma_y(\cdot)$. Then we define the **spectral density function** (or spectrum)

$$s_y(\omega) \equiv \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} \cos(\omega\ell) \Gamma_y(\ell), \quad \omega \in [-\pi, \pi].$$

- Note: $s_y(\omega + 2\pi) = s_y(\omega)$, so only need to consider $[-\pi, \pi]$.
- Since $\Gamma_y(-\ell) = \Gamma_y(\ell)$: $s_y(\omega) = \frac{1}{2\pi} \{ \Gamma_y(0) + 2 \sum_{\ell=1}^{\infty} \cos(\omega\ell) \Gamma_y(\ell) \}$,
and

$$s_y(\omega) = \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} e^{-i\omega\ell} \Gamma_y(\ell), \quad \omega \in [-\pi, \pi],$$

which will be very useful later. Note: $e^{ix} \equiv \cos x + i \sin x$ for $x \in \mathbb{R}$.

- For a white noise process, $s_y(\omega) = \frac{1}{2\pi} \text{Var}(y_t)$ for all $\omega \in [-\pi, \pi]$.

Spectral density function: interpretation

- We can **invert** the spectrum to recover the ACF, cf. slide appendix:

$$\Gamma_Y(\ell) = \int_{-\pi}^{\pi} \cos(\omega\ell) s_Y(\omega) d\omega = \int_{-\pi}^{\pi} e^{i\omega\ell} s_Y(\omega) d\omega.$$

Hence, the spectral density is a function that summarizes everything there is to know about the covariance structure of $\{y_t\}$.

- In particular, $\text{Var}(y_t) = \int_{-\pi}^{\pi} s_Y(\omega) d\omega$.
- We know we may interpret $s_Y(\omega)d\omega$ as the part of the overall variance of y_t attributable to cycles of frequencies $[\omega, \omega + d\omega]$.
- Frequency ω corresponds to $\frac{\omega}{2\pi}$ cycles per period, i.e., cycles of duration $\frac{2\pi}{\omega}$.
- Due to discrete time, can't observe frequencies higher than π . Any such cyclical behavior looks to us exactly like cycles at frequency lower than π . E.g., cycles w. duration 1 period look like a constant to us.

Spectral density function: properties

$$s_y(\omega) \equiv \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} \cos(\omega\ell) \Gamma_y(\ell), \quad \omega \in [-\pi, \pi]$$

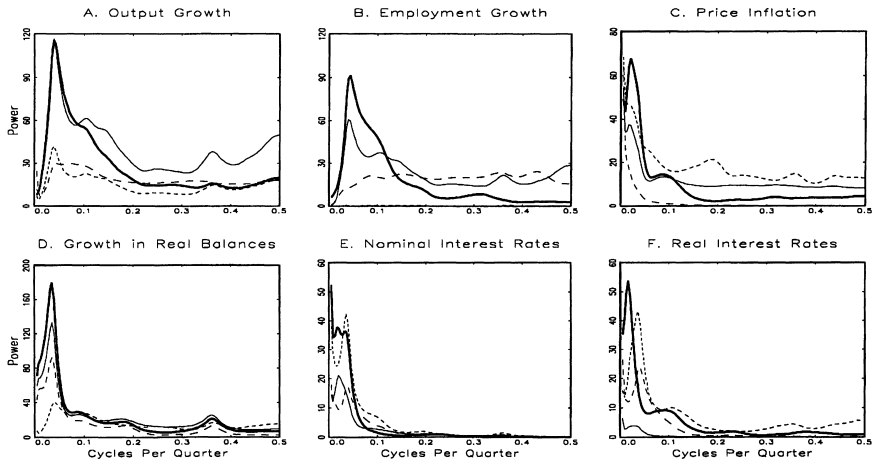
- **Symmetry:** $s_y(-\omega) = s_y(\omega)$ for all $\omega \in [0, \pi]$. Thus, we only need to worry about the value of $s_y(\cdot)$ on $[0, \pi]$.
- **Nonnegativity:** $s_y(\omega) \geq 0$ for all $\omega \in [-\pi, \pi]$. This follows from the positive semidefiniteness of the ACF:

$$\begin{aligned} \sum_{\ell=-\infty}^{\infty} \cos(\omega\ell) \Gamma_y(\ell) &= \lim_{N \rightarrow \infty} \sum_{\ell=-N}^N \left(1 - \frac{|\ell|}{N+1}\right) \cos(\omega\ell) \Gamma_y(\ell) \\ &= \lim_{N \rightarrow \infty} \frac{1}{N+1} \sum_{k=0}^N \sum_{m=0}^N \cos(\omega(k-m)) \Gamma_y(k-m) \\ &= \lim_{N \rightarrow \infty} \frac{1}{N+1} \sum_{k=0}^N \sum_{m=0}^N \{\cos(\omega k) \cos(\omega m) + \sin(\omega k) \sin(\omega m)\} \Gamma_y(k-m) \geq 0. \end{aligned}$$

- An integrable function $s_y(\cdot)$ on $[-\pi, \pi]$ is the spectrum of a cov. stat. process if and only if it is symmetric and nonnegative.

Application: 2nd moments of calibrated macro models

FIGURE 4. — SPECTRA OF DATA AND MODELS



Note: Each panel shows the spectra of postwar quarterly U.S. data (thick line), the real business cycle model (thin line), the sticky-price model (long dashes), and the financial market frictions model (short dashes).

Source: King & Watson (1996). Business cycle frequencies: $[\frac{\pi}{16}, \frac{\pi}{3}]$
(i.e., durations of 6–32 quarters; i.e., 0.03–0.17 cycles per period).

Spectrum of linear filter

Proposition (Brockwell & Davis, 1991, Thm. 4.4.1)

If $\Theta(z) = \sum_{\ell=-\infty}^{\infty} \Theta_{\ell} z^{\ell}$ is abs. summable and $\{x_t\}$ is cov. stat. with spectrum $s_x(\cdot)$, then the spectrum of $y_t = \Theta(L)x_t$ equals

$$s_y(\omega) = |\Theta(e^{-i\omega})|^2 s_x(\omega), \quad \omega \in [-\pi, \pi].$$

- Reminder: $|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$, where $z = a + ib$.
- Calculation is straight-forward:

$$\begin{aligned} s_y(\omega) &= \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} e^{-i\omega\ell} \overbrace{\sum_{k=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \Theta_k \Theta_m \Gamma_x(\ell + m - k)}^{\Gamma_y(\ell)} \\ &= \frac{1}{2\pi} \sum_{\tilde{\ell}=-\infty}^{\infty} e^{-i\omega(\tilde{\ell}+k-m)} \sum_{k=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \Theta_k \Theta_m \Gamma_x(\tilde{\ell}) \\ &= \Theta(e^{-i\omega})\Theta(e^{i\omega}) \frac{1}{2\pi} \sum_{\tilde{\ell}=-\infty}^{\infty} e^{-i\omega\tilde{\ell}} \Gamma_x(\tilde{\ell}). \end{aligned}$$

Spectrum of linear filter (cont.)

$$s_y(\omega) = |\Theta(e^{-i\omega})|^2 s_x(\omega)$$

- The function $\omega \mapsto \Theta(e^{-i\omega})$ is called the **transfer function** of the linear filter.
- $|\Theta(e^{-i\omega})|$ is called the **(amplitude) gain** and its square is the **power transfer function**.
- Can select a transfer function which up/down-weights certain freq's.
- Useful for removing long-run cycles or seasonality.

Spectral density of causal ARMA processes

- Univariate ARMA(p, q):

$$A(L)(y_t - \mu) = \Theta(L)u_t, \quad u_t \sim WN(0, \sigma^2).$$

- Suppose the ARMA is causal. Then

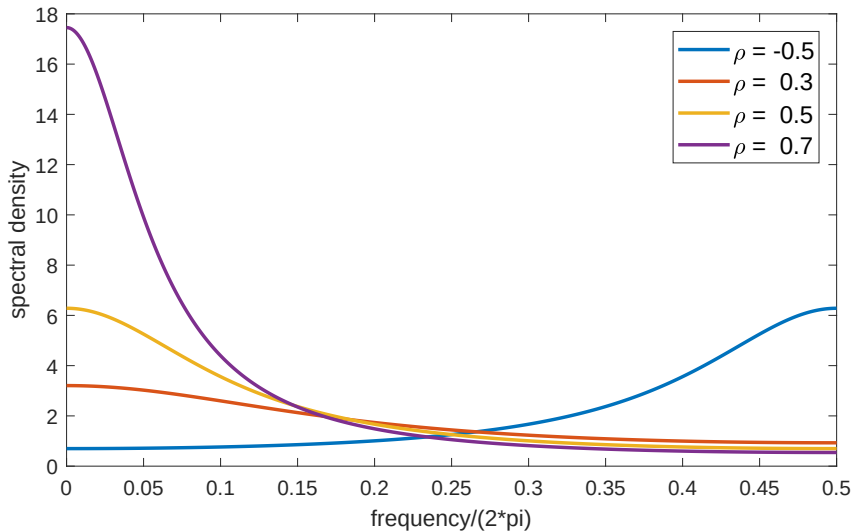
$$y_t = \mu + \Psi(L)u_t, \quad \Psi(L) \equiv A(L)^{-1}\Theta(L).$$

- Since $s_u(\omega) = \frac{\sigma^2}{2\pi}$ for all ω ,

$$s_y(\omega) = |\Psi(e^{-i\omega})|^2 s_u(\omega) = \frac{\sigma^2}{2\pi} \frac{|\Theta(e^{-i\omega})|^2}{|A(e^{-i\omega})|^2} = \frac{\sigma^2}{2\pi} \frac{|1 + \sum_{\ell=1}^q \Theta_\ell e^{-i\omega\ell}|^2}{|1 - \sum_{\ell=1}^p A_\ell e^{-i\omega\ell}|^2}.$$

- Exercise:** For a stable AR(1) $y_t = \rho y_{t-1} + u_t$: $s_y(\omega) = \frac{\sigma^2/(2\pi)}{1 - 2\rho \cos \omega + \rho^2}$.

Example: AR(1) spectrum



Approximation of spectrum by AR/MA

- For large p, q , the class of ARMA spectra is very flexible.
- As the Wold decomp. suggests, we can approximate any cts. spectral density by an invertible MA(q) spectrum, if q is large.

Proposition (Brockwell & Davis, 1991, Cor. 4.4.1)

Let $s(\cdot)$ be a nonnegative, symmetric, continuous function on $[-\pi, \pi]$. Let $\varepsilon > 0$. Then there exists a $q \in \mathbb{N}$ and an invertible MA(q) process

$$y_t = u_t + \sum_{\ell=1}^q \Theta_{\ell} u_{t-\ell}, \quad u_t \sim WN(0, \sigma^2),$$

such that $|s_y(\omega) - s(\omega)| < \varepsilon$ for all $\omega \in [-\pi, \pi]$.

- Since we can go back and forth btw. invertible MA's and causal AR's, the above theorem also holds for approximation by causal AR(p) processes (B&D Cor. 4.4.2).

Kolmogorov-Szegö formula

- The one-step-ahead linear forecast variance of any cov. stat. series $\{y_t\}$ is determined by its ACF, i.e., by its spectrum.
- The spectrum can be approximated by a causal $AR(p)$:

$$A(L)\tilde{Y}_t = u_t, \quad u_t \sim WN(0, \sigma^2).$$

- For any $AR(p)$:
 - ▶ The one-step-ahead linear forecast variance is σ^2 .
 - ▶ $\int_{-\pi}^{\pi} \log s_{\tilde{Y}}(\omega) d\omega = 2\pi \log \frac{\sigma^2}{2\pi} - \underbrace{\int_{-\pi}^{\pi} \log |A(e^{-i\omega})|^2 d\omega}_{=0}$ (cf. B&D p. 191).

Proposition (Brockwell & Davis, 1991, Thm. 5.8.1)

If $\{y_t\}$ is cov. stat., the **Kolmogorov-Szegö formula** applies:

$$\text{Var}^*(y_t \mid \{y_\tau\}_{-\infty < \tau \leq t-1}) = 2\pi \exp \left\{ \frac{1}{2\pi} \int_{-\pi}^{\pi} \log s_y(\omega) d\omega \right\}.$$

Spectral representation theorem

- We now formally justify our interpretation of the spectral density.

Proposition (Brockwell & Davis, 1991, Thm. 4.8.2, Problem 4.25)

If $\{y_t\}$ is cov. stat. with mean zero, then there exist two *orthogonal increment processes* $U(\cdot)$ and $V(\cdot)$ on $[-\pi, \pi]$, which are mutually orthogonal and satisfy $U(-\pi) = V(-\pi) = 0$, such that

$$y_t = \int_{-\pi}^{\pi} \cos(\omega t) dU(\omega) + \int_{-\pi}^{\pi} \sin(\omega t) dV(\omega) \quad \text{for all } t.$$

- An orthog. incr. process $U(\cdot)$ satisfies:
 - i) $E(U(\omega)) = 0$, $\text{Var}(U(\omega)) < \infty$ for all ω .
 - ii) $\text{Cov}(U(\omega_4) - U(\omega_3), U(\omega_2) - U(\omega_1)) = 0$ when $[\omega_1, \omega_2] \cap [\omega_3, \omega_4] = \emptyset$.
 - iii) $\text{Var}(U(\omega + \delta) - U(\omega)) \rightarrow 0$ as $\delta \rightarrow 0$.
- See B&D, ch. 4, for the definition of stochastic integrals. Intuitively,
$$\int_{-\pi}^{\pi} \cos(\omega t) dU(\omega) \approx \sum_{j=0}^{N-1} \cos(\omega_j t) [U(\omega_{j+1}) - U(\omega_j)], \quad \omega_j \equiv \frac{2\pi j}{N} - \pi.$$

Spectral representation: linear filters

- The spectral representation can also be written

$$y_t = \int_{-\pi}^{\pi} e^{i\omega t} dZ(\omega), \quad Z(\omega) \equiv U(\omega) - iV(\omega).$$

- If x_t has spectral representation

$$x_t = \int_{-\pi}^{\pi} e^{i\omega t} dU_x(\omega),$$

and $\Theta(z) = \sum_{\ell=-\infty}^{\infty} \Theta_{\ell} z^{\ell}$ is abs. summ., then (B&D Thm. 4.10.1)

$$y_t \equiv \Theta(L)x_t = \int_{-\pi}^{\pi} e^{i\omega t} \Theta(e^{-i\omega}) dU_x(\omega),$$

i.e.,

$$y_t \approx \sum_{j=0}^{N-1} e^{i\omega_j t} \Theta(e^{-i\omega_j t}) (U_x(\omega_{j+1}) - U_x(\omega_j)).$$

Hence, the filter's transfer function up- or down-weights the cycles at different frequencies. Consistent with spectral density formula.

Spectral distribution function

- We may define the increasing function

$$F_y(\omega) \equiv \text{Var}(U(\omega)) + \text{Var}(V(\omega)), \quad \omega \in [-\pi, \pi].$$

Called the **spectral distribution function**. $F_y(\omega_2) - F_y(\omega_1)$ is fraction of variance attributable to cycles of frequencies $[\omega_1, \omega_2]$.

- Even if ACF is not abs. summ., one can show (using Riemann-Stieltjes integral)

$$\Gamma_y(\ell) = \int_{-\pi}^{\pi} e^{i\omega\ell} dF_y(\omega) \quad \text{for all } \ell \in \mathbb{Z}.$$

- Hence, $F_y(\cdot)$ summarizes all there is to know about the ACF. If $F_y(\cdot)$ is absolutely continuous and the ACF abs. summable, then

$$F_y(\omega) = \int_{-\pi}^{\omega} s_y(\tilde{\omega}) d\tilde{\omega}, \quad \omega \in [-\pi, \pi].$$

So $F_y(\cdot)$ is the “distribution fct” corresponding to “density” $s_y(\cdot)$.

- See slide appendix for $F_y(\cdot)$ with discrete mass points.

Appendix: Random sum of sinusoids calculation

$$\begin{aligned}\frac{1}{2\pi} \sum_{|\ell| < N/2} \cos(\omega_m \ell) \Gamma_y(\ell) &= \frac{1}{2\pi} \sum_{|\ell| < N/2} \cos(\omega_m \ell) \frac{2\pi}{N} \sum_{j=0}^{N-1} s(\omega_j) \cos(\omega_j \ell) \\&= \frac{1}{N} \sum_{j=0}^{N-1} s(\omega_j) \sum_{|\ell| < N/2} \cos(\omega_m \ell) \cos(\omega_j \ell) \\&= \frac{1}{N} \sum_{j=0}^{N-1} s(\omega_j) \sum_{|\ell| < N/2} \frac{\cos((\omega_m - \omega_j)\ell) + \cos((\omega_m + \omega_j)\ell)}{2}. \quad (\dagger)\end{aligned}$$

- On the next slide, we show

$$\sum_{|\ell| < N/2} \cos((\omega_m \pm \omega_j)\ell) = N \times \mathbb{1}(\omega_m \pm \omega_j = 0).$$

- Using symmetry of $s(\cdot)$, the display $(\dagger)$ then becomes (as we wanted)

$$\sum_{j=0}^{N-1} s(\omega_j) \frac{\mathbb{1}(\omega_j = \omega_m) + \mathbb{1}(\omega_j = -\omega_m)}{2} = \frac{s(\omega_m) + s(-\omega_m)}{2} = s(\omega_m).$$

Appendix: Random sum of sinusoids calculation (cont.)

- If $\omega_m \pm \omega_j = 0$, recalling that N is assumed to be odd,

$$\sum_{|\ell| < N/2} \cos((\omega_m \pm \omega_j)\ell) = \sum_{|\ell| < N/2} 1 = \sum_{|\ell| \leq (N-1)/2} 1 = 1 + 2 \frac{N-1}{2} = N.$$

- If $\omega_m \pm \omega_j \neq 0$, **Lagrange's trigonometric identity** states that

$$\sum_{\ell=1}^{(N-1)/2} \cos((\omega_m \pm \omega_j)\ell) = -\frac{1}{2} + \frac{\sin\left(\left(\frac{N-1}{2} + \frac{1}{2}\right)(\omega_m \pm \omega_j)\right)}{2 \sin((\omega_m \pm \omega_j)/2)}.$$

The numerator in the last fraction above is zero, since

$$\left| \sin\left(\frac{N}{2} \left(\frac{2\pi(m \pm j)}{N} - (\pi \pm \pi)\right)\right) \right| = |\sin(\pi(m \pm j))| = 0.$$

Hence,

$$\sum_{|\ell| < N/2} \cos((\omega_m \pm \omega_j)\ell) = 1 + 2 \sum_{\ell=1}^{(N-1)/2} \cos((\omega_m \pm \omega_j)\ell) = 0.$$

Appendix: Inverting the spectrum

$$\begin{aligned}s_y(\omega) &\equiv \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} e^{-i\omega\ell} \Gamma_y(\ell) \\ \implies \int_{-\pi}^{\pi} e^{i\omega m} s_y(\omega) d\omega &= \int_{-\pi}^{\pi} e^{i\omega m} \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} e^{-i\omega\ell} \Gamma_y(\ell) d\omega \\ &= \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} \Gamma_y(\ell) \int_{-\pi}^{\pi} e^{i\omega(m-\ell)} d\omega \\ &= \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} \Gamma_y(\ell) 2\pi \mathbb{1}(\ell = m) \\ &= \Gamma_y(m).\end{aligned}$$

- The first equality uses Fubini and abs. summability of the ACF.
- The third equality uses $\int_{-\pi}^{\pi} e^{i\omega k} d\omega = 2\pi$ for $k = 0$ and

$$\int_{-\pi}^{\pi} e^{i\omega k} d\omega = \int_{-\pi}^{\pi} \cos(\omega k) d\omega + i \int_{-\pi}^{\pi} \sin(\omega k) d\omega = 0 \quad \text{for } k \in \mathbb{Z} \setminus \{0\}.$$

Appendix: Non-continuous spectral distr. function

- If the Wold decomposition of $\{y_t\}$ has a deterministic component, then the spectral distr. fct $F_y(\cdot)$ is not absolutely continuous (i.e., it has jumps due to point masses).
- The abs. cts part of $F_y(\cdot)$ (subtracting out the jumps) is then the spectral distr. fct. of the $MA(\infty)$ component in the Wold decomp.
- Since we typically work with purely linearly nondeterministic series, we don't have to worry about spectral distr. fcts that are not abs. cts. Thus, as long as the ACF is abs. summ., $s_y(\cdot)$ provides the same information as $F_y(\cdot)$.
- Regardless, for forecasting purposes it is sufficient to know $s_y(\cdot)$, since a deterministic term does not contribute to forecast errors.
- Details in B&D, ch. 5.6–5.8.